

Cheung Chung Ming

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Education

University of Southern California <i>Ph.D. in Computer Science (GPA: 4.00 / 4.00)</i> Thesis: Data-Driven Methods for Increasing Real-Time Observability in Smart Distribution Grids	2014 - 2021 United States
University of Hong Kong <i>Bachelor of Engineering (Computer Engineering) (GPA: 4.10 / 4.30)</i>	2011 - 2014 Hong Kong

Work Experience

NVIDIA <i>Inference Software Engineer, TensorRT (Senior Software Engineer)</i> - Core member of TensorRT frontend team, which was responsible for optimization of models such as layer fusions, handling of dynamic shapes, autotuning format and data precision, and bridge to backend libraries. - Experienced in investigating performance issues, accuracy gap, and memory utilization of models post-optimization. Utilizing tools such as polygraphy for runtime testing, ONNX model manipulations, and Nsight for inspecting CPU/GPU workloads. - Collaborated closely with NVIDIA Deep Learning Accelerator (DLA) compiler team to build a robust and optimal bridge between TensorRT and the DLA backend. Projects include bringing up DLA on new hardware and removing static dependency link between TensorRT and DLA. - Mentored a new team member to work on removing synchronous CUDA calls from the TensorRT core library, which resulted in better overall GPU utilization. Scoped the task, debugged tricky issues from asynchronous CUDA calls and planned the deployment. - Implemented new features in the TensorRT library, including "debug tensors" feature for developers to inspect intermediate tensor values and "strongly-typed" feature to ensure TensorRT follows data type designated by the input model.	2021 - Present United States
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Technical Skills

Languages: C++, Python
Technologies: TensorRT, polygraphy, Nsight, ONNX, CUDA, PyTorch, gitlab, docker, numpy, AI-assisted coding (Cursor, ClaudeCode), scikit-learn, pandas, matplotlib, cvxpy, PI System
Expertise: Inference optimization, Machine learning, Data processing

Research Experience

List of publications can be found at ccming3.github.io/publications/

PhD Thesis: Data-Driven Methods for Increasing Real-Time Observability in Smart Distribution Grids - Investigated the use of grid topology information as spatial features input for improving load forecasting results by learning dependencies between spatial features in time series in smart grids. - Demonstrated enhancements in times series analytics in smart grids from increased observability using the aforementioned results. Technologies involved: Convolutional Neural Networks, Graph Convolutional Networks, Convex Optimization, Time Series Clustering, Feature Extraction
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DEEPSOLAR

- Developed an Ensemble Learning model for Short Term Load Forecasting.
- Developed a data-driven model for accurate disaggregation of hidden solar energy behind meters from AMI measurements.
- Technologies involved:** Ensemble Learning, Convex Optimization, Time Series Clustering

CiSoft (Center for Smart Oilfield Technology) Projects

- Used machine learning models to handle large industrial datasets to solve time series analytics.
- Technologies involved:** Ontology, Convolutional Neural Networks, Auto-encoders